Abstract*— *This project develops and evaluates regression models using the provided dataset. The aim here is to predict the target variable using regression techniques. The process includes data visualization, preprocessing, scaling, model training, and performance evaluation. Multiple regression models are compared to find the best-performing model.

INTRODUCTION

Regression models are used in machine learning to predict continuous values based on input data. In this coursework, several regression techniques were trained and compared. The aim was to understand the relationships in the dataset and build a model that accurately predicts outputs without overfitting.

METHODOLOGY

The dataset was first imported into Python using the Pandas library. Basic analysis and visualization was done using correlation heatmap which helps us understand the relationships between the variables. All columns except the target variable **y** were used as input features.

The dataset was divided into training data (80%) and testing data (20%) to evaluate performance on unseen data. StandardScaling was used here so that all the data is equally scaled and perform while giving the same importance.

The following regression models were used:

- Linear Regression
- Ridge Regression
- Lasso Regression
- ElasticNet Regression
- Polynomial Regression
- Random Forest Regression

Pipelines were used to combine scaling and training steps properly and avoid data leakage. Each Model's performance was evaluated by Mean Square & Root Mean Squared Error i.e.,(MSE & RMSE) and R^2 score.

RESULTS AND OUTPUTS

Model performance was evaluated using Mean Squared Error, Root Mean Squared Error, and R^2 score to calculate the predictive ability.

MODEL	MSE	RMSE	R2
Linear Regression	7.462779	2.731809	0.882120
Lasso Regression	7.573210	2.751947	0.880376
Ridge Regression	7.587227	2.754492	0.880154
ElasticNet	7.699329	2.774766	0.878384
Polynomial Regression	6831.79	82.6547	-106.91
Random Forest	11.194	3.345	0.823171

1. Linear Regression produced the lowest error and the highest R^2 score among the models. Whereas Ridge, Lasso, and ElasticNet got similar result but were slightly lower.
2. Polynomial Regression performed very poorly because it overfitted the training data and failed to predict correctly on the test data.
3. Random Forest was also tested, but the simpler regression models line linear regression already showed a strong performance.
4. Scatter plots which compared the predicted and actual values showed that Linear Regression predictions were closely matching with the real values, which means, it has great performance.

CONCLUSION

1) This project shows that regression models can predict numerical values well when proper preprocessing and feature scaling are applied. Linear Regression performed best, showing that the dataset mainly follows linear relationships.

2) More complex models, like Polynomial Regression, caused overfitting and reduced performance. Random Forest was also tested, but simpler regression models already gave strong results. Future improvements could include feature engineering and parameter tuning to further improve accuracy.

REFERENCES

[1] G. James et al., An Introduction to Statistical Learning, Springer,2013.
[2] Scikit-learn Documentation, https://scikit-learn.org